Dominik Wüst

M.Sc. Computer Science – Machine Learning & Rendering Engineer

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PROFILE

Computer Science graduate specializing in real-time computer graphics, GPU programming and machine learning. Experienced in C++, Vulkan and Python through research at KIT and Fraunhofer IOSB. Interested in neural rendering, differentiable graphics and computer vision.

EXPERIENCE

Oct 2021 – Mar 2026
Research Assistant | Fraunhofer IOSB
Karlsruhe, Germany

- Applied pretrained computer vision models for object detection, segmentation, image retrieval, and vision-language embeddings to accelerate Structure-from-Motion pipelines
- Deployed deep learning models in C++ using ONNX Runtime and clip.cpp, converting models to ONNX and modifying computation graphs to support dynamic batch sizes and image resolutions
- Developed and shipped cross-platform software and maintained CI/CD pipelines using GitLab and Docker

C++ | Qt | OpenCV | ONNX Runtime

May 2025 – Aug 2025
Research Assistant | Karlsruhe Institute of Technology (IVD)
Karlsruhe, Germany

- Contributed to a LLM-based shader optimization pipeline by integrating the Falcor renderer and adding support for cooperative vector acceleration
- Designed related LLM prompts and analyzed proposed optimizations automatically
- Co-author on related paper under submission

Python | C++ | Slang

Nov 2020 – Jul 2021
Student Assistant | Karlsruhe Institute of Technology (CAPP)
Karlsruhe, Germany

- Tutored a group of students in computer engineering
- Courses in fundamentals of microelectronics, design and construction of logic circuits

LaTeX | C | Presentations

EDUCATION

2023 – 2026
M.Sc. Computer Science | Karlsruhe Institute of Technology (KIT)
Thesis: Automatic Shader Function Level of Detail using Neural Networks and Symbolic Regression – Preliminary Grade 1.2
Computer Graphics | Anthropomatics and Cognitive Systems

2018 – 2023
B.Sc. Computer Science | Karlsruhe Institute of Technology (KIT)
Thesis: Real-Time Neural Path-Guiding – Minor: Electrical Engineering – Grade 2.1

2015 – 2018
Allgemeine Hochschulreife | Technisches Gymnasium Schwäbisch Hall
Specialization: Computer Science – Grade 1.2

SKILLS

Programming

C++ professional
Python professional
C# personal projects
Rust personal projects

Graphics & GPU
Vulkan | Slang | GLSL | CUDA

Machine Learning
PyTorch | ONNX Runtime | tiny-cuda-nn

Computer Vision
OpenCV | OpenMMLab | clip.cpp

Frameworks & UI
Qt | HTML/CSS | ImGui | React

Tooling
Docker | CMake | GitLab CI/CD | Claude Code

LANGUAGES

German native
English fluent

INTERESTS

Weight Lifting | Hiking | PC Modding | Spaceflight

PROJECTS

2026

Efficient Validation for LLM-Generated Rendering Optimizations

M. Dereviannykh, D. Klepikov, H. Brucker, **D. Wüst**, B. Krimmel, R. Dolp, C. Dachsbacher

`renderopt.github.io` *paper coming soon*



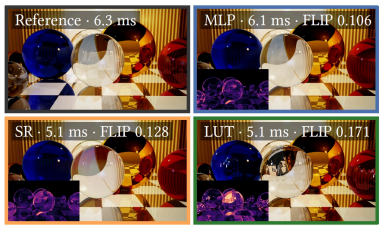
LLM-driven evolutionary search finds shader and render-pass speedups of up to 2.19×; a Bayesian sequential test proves them visually safe with 47–85% fewer validation renders.

LLM · Python · GLSL

2026

Automatic Shader Function Level of Detail using Neural Networks and Symbolic Regression

Code will be published after grading



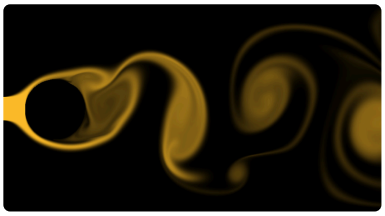
Master's thesis: a data-driven approach to shader level of detail. Tiny MLPs and symbolic expressions are fitted automatically to approximate shader functions. The pipeline collects training data, fits the models, optimizes hyperparameters and evaluates candidates by rendering the shader end-to-end. It also explores fitting the models with respect to the final image using differentiable rendering.

Machine Learning · Python · Slang

2023

2D Fluid Simulation on the GPU

`github.com/dom-wuest/FluidSim2D`



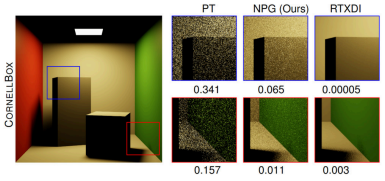
An incompressible fluid simulation built on Vulkan compute shaders. It operates on a two-dimensional grid and supports solid and fluid cells with various initial conditions. Performance bottlenecks were analyzed with NVIDIA Nsight Graphics and optimized through ghost zoning, kernel decomposition and loop unrolling.

C++ · Vulkan · CFD

2023

Real-Time Neural Path-Guiding

`github.com/dom-wuest/NeuralPathGuiding`



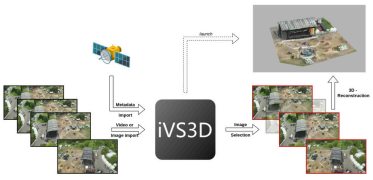
Bachelor's thesis: a compact MLP predicts the incident illumination at surfaces and guides subsequent rays. Training data is collected while rendering and the network is trained online to predict vMF distributions. In the next frame these steer the rays at the first visible surface towards bright regions.

Path-Tracing · Machine Learning · CUDA

2021 *Present*

Intelligent Video Sampler 3D

`github.com/iVS3D/iVS3D`



Preprocessing of image sequences for Structure-from-Motion using computer vision algorithms and deep learning models. The open-source project has been developed actively for over five years and runs on Windows and Linux. A plugin architecture selects frames by camera movement, deep visual similarity and GPS position, and masks dynamic objects such as vehicles automatically before a configurable SfM workflow starts.

C++ · Qt · OpenCV